

Modern CFD Algorithms for the Cone and Shock Tube

*Finite-volume fluxes, high-order reconstruction
and verifiable numerical comparisons*

A teaching companion to the 2026 Python notebooks

English edition · 08 October 2026

The question this lecture answers

How do numerical methods move mass, momentum and energy across a cell face, and why do their computed shocks and contacts differ?

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- 15 implemented Euler face fluxes + standalone JST.
- 20 finite-volume reconstructions and DG degrees 0, 1 and 2.
- Algorithm steps, advantages, limitations and coding exercises.
- Actual Sod results, fixed uniform-grid refinement and exact-reference diagnostics.

Scope

This lecture teaches the algorithms selectable in the shared cone/shock-tube code. The plotted benchmark solves inviscid planar Euler equations. It does not validate physical viscous transport or cover every CFD method in use.

Source baseline: public repository commit `e7cd1655a07cbae51ab98d22bec87c274393fa55`. Numerical algorithms and recorded results are unchanged by this lecture.

01 / TEACHING ROUTE

How to teach and use these notes

Session	Topics	Notebook activity
1: equations and updates	Euler waves, cell means, conservation, SSPRK3	Read Sod profiles; implement a local LF flux.
2: face solvers	Roe, HLL/HLLC, splitting, AUSM, entropy/JST	Hold CWENO3 fixed and compare fluxes.
3: high-order spatial methods	MUSCL, CWENO, WENO, TENO, BVD, DG	Hold HLLC fixed and compare reconstructions.
4: evidence and decisions	Admissibility, grid errors, interface width, failure labels	Compare six uniform grids and explain a method choice.

Learning outcomes

- Trace a conservative update from reconstructed states to one shared flux.
- Distinguish flux, reconstruction, time integration and physical models.
- Explain why smooth formal order does not predict shock-profile accuracy.
- Compare errors, positivity, conservation and wave resolution together.

Reading route

Foundations → face fluxes → reconstruction → DG and safeguards → measured comparisons → laboratory and exercises. Each named method has an algorithm, an advantage and a limitation.

Prerequisites

Conservation laws, ideal-gas pressure, basic eigenvectors and Python arrays. The finite-volume update is developed here; no packaged CFD solver is required.

02 / GOVERNING EQUATIONS

One physical system, many algorithms

Key equation

$$U = (\rho, \rho u, E)^T$$

$$F(U) = (\rho u, \rho u^2 + p, u(E+p))^T$$

$$\partial U / \partial t + \partial F / \partial x = 0$$

Key equation

$$p = (\gamma - 1)[E - \rho u^2 / 2], \quad a = \sqrt{(\gamma p / \rho)}, \quad H = (E + p) / \rho$$

- ρ is density; u is normal velocity; E is total energy per volume.
- Pressure closes the ideal-gas equations; a sets an acoustic-speed scale.
- Euler wave speeds are $u-a$, u and $u+a$ in planar flow.
- The shared code carries transverse momentum too; it is zero in Sod.

Physical versus numerical dissipation

The Euler comparisons have no molecular-viscosity term. Damping from LF, JST or a limiter is numerical dissipation. A new flux does not add physical viscosity or a turbulence model.

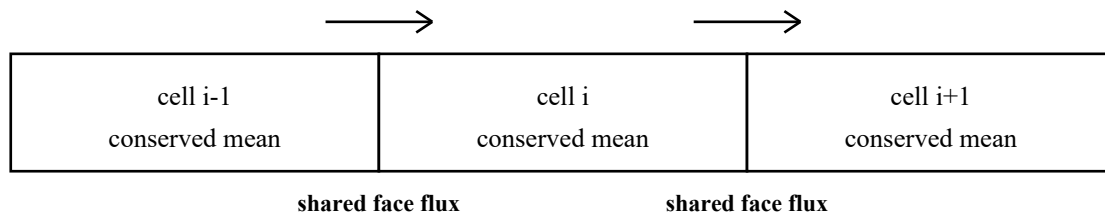
Sod initial data

Region	ρ	u	p
Left of $x=0.5$	1	0	1
Right of $x=0.5$	0.125	0	0.1

Domain $[0,1]$; $\gamma=1.4$; requested time $t=0.2$; outflow boundaries.

03 / FINITE VOLUME

Build a conservative cell update

**Key equation**

$$\frac{dU_i}{dt} = -(F_{i+1/2} - F_{i-1/2})/\Delta x$$

- Integrate the conservation law over a cell and store its conserved mean.
- Reconstruction supplies two states at each face.
- The face solver turns these two states into one flux.
- Use the identical flux in both neighbors. Interior contributions cancel in the domain budget.

A small mass-balance calculation

Let $\Delta x=0.01$, $\Delta t=0.001$, left mass flux=0.20 and right mass flux=0.25. The Euler substep changes density by $-(0.001/0.01)(0.25-0.20)=-0.005$. A density of 1 becomes 0.995. After summing cells, only boundary flux contributions remain.

Coding check

Compute each face flux once into an array. Check a two-cell update by hand. Updating primitive variables independently generally does not reproduce the conservative Euler update.

04 / WAVE PHYSICS

What must a shock tube resolve?

Feature	Physical signature	Numerical challenge
Rarefaction	Continuous expansion; density and pressure vary smoothly.	Retain smooth accuracy without flattening the fan.
Contact	Density jump; pressure and normal velocity continuous.	Limit spreading of a transported density interface.
Shock	Compression jump in density, pressure and velocity.	Capture strength and position without ringing.

Why use cone flow and shock tubes?

Cone flow couples wave capturing to geometry, source integration, boundaries and a structured spatial flow. A planar shock tube removes the cone source and isolates transient wave interactions. These are complementary tests that stress different components.

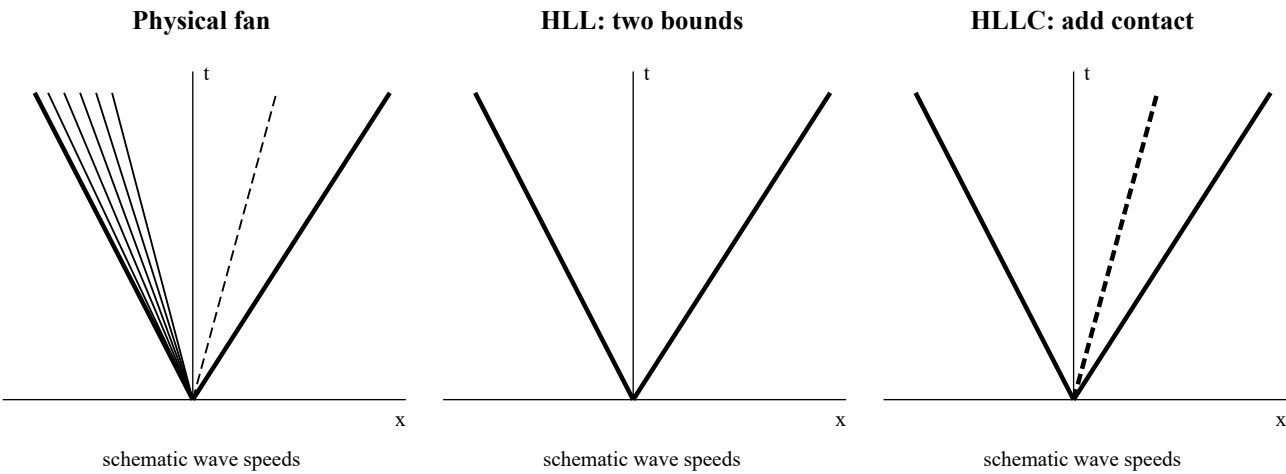
Interpret a rounded transition

A finite-volume jump spans finite cells. Refinement should reduce its physical thickness. A rarefaction is physically smooth and should not be sharpened into a step.

- Sod: expansion, contact and shock.
- Lax: stronger waves.
- Double rarefaction: low-pressure/density stress.
- Stationary contact: contact diffusion without transport.

04B / VISUAL REASONING

From a wave fan to a face flux



At a local face, left/right states define a Riemann problem. Rays have slope x/t equal to a wave speed. The flux samples the state at the face ray $x/t=0$. This original schematic shows topology, not the measured Sod wave speeds.

Description	Retained information
Exact local solver	Shock/rarefaction branches and a contact; nonlinear fan structure.
HLL approximation	Two outer bounds and one intermediate state; contact not represented separately.
HLLC approximation	Outer bounds plus a contact dividing two star states.

Classroom prompt

Where is $x/t=0$ in each panel? If the entire fan travels right, which physical state supplies the face flux?
Answer: the left state. Repeat with an entirely left-going fan.

Teaching idea: use local wave diagrams before full numerical profiles, following Ketcheson, LeVeque and del Razo, Riemann Problems and Jupyter Solutions, Euler approximate-solvers chapter. See the linked reading map at the end.

05 / TIME INTEGRATION

Advance with SSPRK3

Key equation

$$A = U^n + \Delta t L(U^n)$$

$$B = 3U^n/4 + [A + \Delta t L(A)]/4$$

$$U^{n+1} = U^n/3 + 2[B + \Delta t L(B)]/3$$

Key equation

$$\Delta t_{FV} = CFL \cdot \Delta x / \max(|u|+a)$$

- Recompute the full spatial operator at each stage, including boundaries.
- Clip Δt to reach the requested time. Recorded uniform-grid CFL is 0.25.
- Reject invalid stage means and retry at a smaller step.
- Strong-stability-preserving terminology does not guarantee positivity or no oscillations for every spatial method.

Worked estimate

For $\Delta x=0.01$, $\max(|u|+a)=2$ and $CFL=0.25$, $\Delta t=0.00125$. Halving Δx halves the allowable step. Refinement increases both cell work and step count.

Step pseudocode: choose $\Delta t \rightarrow$ reconstruct admissible states $\rightarrow$ evaluate fluxes $\rightarrow$ SSPRK stage $\rightarrow$ validate means $\rightarrow$ repeat $\rightarrow$ accept, or restore the old state and halve Δt . Retry counts are finite.

Exact and linearized local wave fans

godunov - exact Riemann

Algorithm. Solve the ideal-gas star-pressure equation by bracketing and bisection; construct shock/rarefaction/contact waves; sample at $x/t=0$ and evaluate its physical flux. Includes a vacuum branch.

Advantage. A direct benchmark for approximate fluxes; retains the complete local wave structure.

Limitation. Exact **local** Riemann flux; the complete spatial/time discretization still has error. More expensive than approximate solvers.

roe - Roe with entropy correction

Algorithm. Form square-root-density-weighted velocity/enthalpy; decompose the state jump into acoustic, contact and shear eigenwaves; subtract half the absolute-wave-speed weighted jump from the mean physical flux. Smooth small acoustic eigenvalues with a Harten-style entropy correction.

Advantage. A compact wave-by-wave treatment with explicit contact and acoustic information.

Limitation. Resolves characteristic waves efficiently. The entropy correction is not a proof of full-scheme entropy stability; positivity is not automatic.

roe-nc - uncorrected Roe

Algorithm. Use the same Roe decomposition without the small-eigenvalue entropy correction.

Advantage. Isolates the effect of Roe entropy correction in a diagnostic experiment.

Limitation. Omitting the entropy correction can admit nonphysical expansion behavior; this diagnostic is not a general replacement for corrected Roe.

Roe dissipation

$$\hat{F} = (F_L + F_R)/2 - \frac{1}{2} \sum |\lambda_k| \alpha_k r_k$$

Decompose the jump into waves r_k ; their speed magnitudes set wave-dependent dissipation.

07 / HLL FAMILY

Two waves, then restore the contact

hlle - two-wave HLL

Algorithm. Bound the fan with $S_L = \min(u_L - a_L, u_R - a_R, 0)$ and $S_R = \max(u_L + a_L, u_R + a_R, 0)$; use the HLL intermediate flux between these bounds and the upwind physical flux outside them.

Advantage. Few intermediate-state calculations; useful as the implemented HLLC fallback.

Limitation. This implementation uses **Davis acoustic bounds**, despite the hlle key. Omitting the contact wave increases contact diffusion.

hllc - contact-restoring HLL

Algorithm. Estimate the two outer wave speeds and the contact speed; construct left/right star states; select the flux by the wave-speed signs. Fall back locally to HLL when star density/pressure is invalid or nonfinite.

Advantage. Restores the contact wave without an iterative exact star-pressure solve.

Limitation. Better contact resolution than two-wave HLL; moving contacts still spread on finite grids.

Key equation

$$S_L = \min(u_L - a_L, u_R - a_R, 0)$$

$$S_R = \max(u_L + a_L, u_R + a_R, 0)$$

$$F_{\text{HLL}} = [S_R F_L - S_L F_R + S_L S_R (U_R - U_L)] / (S_R - S_L)$$

Name-to-code caution

The hlle key uses Davis min/max acoustic bounds, not Einfeldt/Roe-averaged bounds. The label alone does not specify the implemented wave-speed estimate.

08 / SIMPLER UPWINDING

LF and flux-vector splitting

rusanov - local Lax-Friedrichs

Algorithm. Use $F_{\text{hat}} = (F_L + F_R)/2 - \alpha(U_R - U_L)/2$, with $\alpha = \max(|u_L| + a_L, |u_R| + a_R)$ at each face.

Advantage. Short, transparent implementation; useful for debugging and comparison.

Limitation. Simple acoustic-speed dissipation; generally more diffusive on slow/contact waves.

global-lf - global LF flux

Algorithm. Use the same central-minus-jump formula with a **single maximum speed over all faces in the current residual evaluation**.

Advantage. A single speed makes the chosen dissipation easy to inspect.

Limitation. A fast region sets damping everywhere, increasing diffusion in slow regions. This is the semidiscrete flux, not a classical fully discrete LF update.

vanleer - Van Leer flux-vector splitting

Algorithm. Split each state's flux into positive/negative parts using Mach-number polynomials for subsonic flow; add $F^+(U_L) + F^-(U_R)$. Supersonic states contribute only in their upwind direction.

Advantage. Directional physical fluxes with a smooth subsonic-to-supersonic construction.

Limitation. Smooth sonic splitting, with contact diffusion. This is distinct from the Van Leer **slope limiter**.

steger-warming - eigenvalue flux splitting

Algorithm. Split the physical-flux eigenvalues as $\lambda^\pm = (\lambda \pm |\lambda|)/2$ for each state; combine the left positive and right negative flux contributions.

Advantage. A systematic eigenvalue split provides a clear upwind construction.

Limitation. Straightforward upwind splitting; can diffuse contacts.

Separate mass transport and pressure

Key equation

$$\mathbf{F}_{\text{hat}} = \dot{m} \cdot (1, u_{\text{up}}, H_{\text{up}})^T + (0, p_{\text{face}}, 0)^T$$

The mass-flux sign chooses upwind velocity and enthalpy. Split interface pressure acts on normal momentum. The four-component code also transports transverse momentum.

ausm - original AUSM

Algorithm. Split mass advection by left/right local Mach polynomials and split pressure separately. Upwind velocity and enthalpy according to the interface mass-flux sign; add face pressure to normal momentum flux.

Advantage. Separates mass transport from pressure, making family changes easy to trace.

Limitation. The original split lacks the dedicated pressure/velocity corrections of AUSM+-up; all-Mach behavior needs separate tests.

ausm-plus - AUSM+

Algorithm. Use an enthalpy-based interface sound speed and improved Mach/pressure polynomials, then the same mass-advection/pressure construction.

Advantage. Improves Mach/pressure splitting without a local iterative Riemann solve.

Limitation. Improved splitting around sonic transitions; no explicit AUSM+-up low-Mach corrections.

What changes within the family?

Interface sound speed, Mach/pressure polynomials and dissipation corrections. Conserved variables, equation of state and physical Euler flux remain the same.

AUSM+-up, AUSM+-up2 and SLAU2

ausm-up - AUSM+-up

Algorithm. Add a Mach-regulated pressure-jump correction to interface mass flux and a velocity-jump correction to interface pressure. The code uses reference Mach **0.1**.

Advantage. Adds explicit pressure/velocity coupling for tests across Mach regimes.

Limitation. Designed to improve pressure/velocity coupling across Mach regimes. Performance depends on the correction parameters and test.

ausm-up2 - AUSM+-up2

Algorithm. Keep the pressure-jump correction in the mass flux; replace the interface-pressure expression with a velocity-magnitude-based dissipation term. The implemented reference Mach remains **0.1**.

Advantage. Tests another pressure-dissipation mechanism with the same mass-correction backbone.

Limitation. Pressure dissipation and the reference-Mach setting can influence diffusion. The Sod study does not establish superiority across Mach regimes.

slau2 - SLAU2

Algorithm. Use arithmetic mean sound speed, density-weighted velocity splitting and a low-Mach pressure-jump correction for mass flux; use velocity-magnitude-based interface pressure dissipation.

Advantage. A distinct all-speed splitting design for controlled comparisons.

Limitation. Changing the dissipation design does not guarantee smaller errors in every flow. These shock-tube runs do not validate a low-Mach asymptotic limit.

Mechanism before ranking

Pressure-jump corrections change mass/pressure coupling; velocity terms change pressure dissipation. All-speed intent does not imply the smallest Sod error. Hold reference-Mach settings fixed in grid comparisons.

Entropy core and JST dissipation

ec-lf - entropy-conservative core + LF

Algorithm. Construct a Chandrashekar-style central flux using logarithmic means of density and $\beta = \rho/(2p)$; add local LF jump dissipation.

Advantage. Separates an entropy-conservative central construction from added dissipation.

Limitation. LF dissipation can spread contacts. A checked central face entropy identity is not a proof for the full reconstructed and time-integrated scheme.

jst - standalone central stencil

Algorithm. Evaluate F at the averaged face state; subtract speed-scaled second/fourth-difference dissipation. Pressure curvature activates stronger second-difference damping near shocks.

Advantage. The sensor exposes the damping mechanism explicitly.

Limitation. The current stencil can ring near jumps. It is numerical damping, not physical viscosity, and is not interchangeable with DG or every reconstruction.

Key equation

$$\varepsilon_2 = 0.5 s_p, \quad \varepsilon_4 = \max(0, 0.02 - \varepsilon_2)$$

Different entropy claims

A central face identity, dissipative two-state flux, reconstructed semidiscrete scheme and fully discrete boundary-treated scheme are distinct claims. Face checks do not prove global entropy stability.

12 / HIGH-ORDER RECONSTRUCTION

Cell means are not nodal samples

Key equation

$$U_i = (1/\Delta x) \int_{\text{cell } i} P_i(x) dx$$

$$U_{i+1/2} = P_i(x_{i+1/2})$$

- Fit moments to cell means; evaluate the fitted representation at faces.
- A limiter or nonlinear weight suppresses candidates that cross a jump poorly.
- Higher formal order requires smooth information; a wider stencil does not make a discontinuity smooth.
- Characteristic reconstruction projects into local wave variables, reconstructs and transforms back.

Three independent controls

Flux: interface dissipation. Reconstruction: face data. Mesh spacing: physical resolution. Changing all three at once obscures why an answer changes.

A fair experiment

Hold HLLC, grid, boundaries, final time and SSPRK3 fixed. Change only reconstruction. Inspect contact width, overshoot and density/pressure errors, then repeat matched grids before claiming observed order.

13 / LINEAR BASELINES

First order, MC and minmod

first

Algorithm. Keep each cell's mean constant; its two face states equal that mean.

Advantage. A minimal conservative baseline for verifying higher-order modifications.

Limitation. First order, inexpensive and diffusive.

muscl

Algorithm. Fit a linear polynomial with the monotonized-central (MC) slope: limit the centered slope by twice either one-sided difference; set it to zero when the differences disagree in sign.

Advantage. A compact second-order stencil balancing centered slopes and jump control.

Limitation. Second order in smooth regions; reduces slopes around jumps/extrema.

muscl-minmod

Algorithm. Use the smaller magnitude of the two same-sign one-sided slopes, or zero when signs differ.

Advantage. Simple, cautious slopes that are easy to verify by hand.

Limitation. More conservative limiting, with greater diffusion.

Hand calculation

Means 1.0, 1.2, 1.3 give $a=0.2$, $b=0.1$. MC slope= $\min(2a, 2b, (a+b)/2)=0.15$; minmod=0.1. On normalized coordinates $[-\frac{1}{2}, \frac{1}{2}]$, MC face states are 1.125 and 1.275; minmod gives 1.15 and 1.25. Both preserve mean 1.2.

14 / MUSCL CHOICES

Smoothness versus compression

muscl-vanleer

Algorithm. Use the harmonic slope $2ab/(a+b)$ for same-sign differences a, b , otherwise zero.

Advantage. A smooth harmonic transition between permitted slopes.

Limitation. Remains second order in smooth regions and flattens at extrema under the implemented same-sign condition. Contacts can still spread.

muscl-superbee

Algorithm. For same-sign differences, choose the larger of the two bounded slopes $\min(2|a|, |b|)$ and $\min(|a|, 2|b|)$, retaining their sign.

Advantage. Can keep monotone interfaces more compact than gentler limiters.

Limitation. Sharper, more compressive profiles; not universally the most accurate choice.

muscl-vanalbada

Algorithm. Use the smooth slope $ab(a+b)/(a^2+b^2+\epsilon)$ for same-sign differences, otherwise zero.

Advantage. A smooth rational response without abrupt one-sided slope selection.

Limitation. Remains second order and returns zero at opposing slope signs; smooth extrema and transported contacts can lose resolution.

Same data, different slopes

For $a=0.2$, $b=0.1$: Van Leer=0.13333; Superbee=0.2; Van Albada $\approx$ 0.12. When signs disagree, these implemented variants return zero. Sharper permitted slopes can be more compressive.

15 / CANDIDATES

ENO and cell-based CWENO

eno2

Algorithm. Choose the one-sided linear stencil with the smaller neighboring state jump, component by component.

Advantage. Avoids the larger local jump by choosing the less troubled stencil.

Limitation. Second-order smooth reconstruction; abrupt stencil selection can affect profiles.

cweno3

Algorithm. Build a third-order optimal polynomial from cell means, two lower-order candidates and a central candidate; combine them with smoothness-dependent nonlinear weights.

Advantage. One cell polynomial supports faces and geometric-source integration.

Limitation. Candidate fitting and nonlinear weighting cost more than a linear limiter; shocks and admissibility scaling reduce the smooth formal order.

cweno5

Algorithm. Apply the same central-WENO construction with a fifth-order optimal polynomial and wider candidates.

Advantage. Higher smooth accuracy while retaining a cell-based polynomial.

Limitation. Higher smooth accuracy, more stencil work; limiting can reduce order.

Key equation

$$\omega_k = \alpha_k / \sum_j \alpha_j, \quad \alpha_k = d_k / (\beta_k + \epsilon)^2$$

d are optimal weights; β measure derivative smoothness. A central CWENO candidate allows the linear combination to recover the optimal wide-stencil polynomial.

Weight waves instead of components

cweno3-char

Algorithm. Project CWENO3 candidates into the local Euler eigenbasis; compute nonlinear weights per characteristic wave; transform back.

Advantage. Weights individual waves instead of mixing conservative components.

Limitation. Limits coupling between different wave families, with extra eigenbasis work.

cweno5-char

Algorithm. Use characteristic weighting with the fifth-order CWENO construction.

Advantage. Combines wave weighting with a higher-order smooth polynomial.

Limitation. A wider stencil and eigenbasis work increase cost; limiting near jumps can remove the formal higher-order advantage.

- Evaluate a local Euler eigenbasis at the cell state.
- Project candidate coefficients into wave variables.
- Compute nonlinear weights for each wave component.
- Transform back and apply admissibility controls.

Advantage and cost

Separate wave weighting can reduce acoustic/contact coupling. It requires eigenvectors and extra transformations. It alone guarantees neither positive pressure nor no overshoot.

17 / WENO WEIGHTING

Jiang-Shu versus Z weights

weno3-js

Algorithm. Form characteristic face values on small FV candidate stencils; blend using Jiang-Shu weights $\alpha_k = d_k / (\beta_k + \epsilon)^2$.

Advantage. A small characteristic face stencil with nonlinear smoothness weighting.

Limitation. Third-order smooth faces; nonlinear weights can lose accuracy near smooth critical points.

weno3-z

Algorithm. Use the same third-order candidates with Z weights $\alpha_k = d_k [1 + (\tau / (\beta_k + \epsilon))^2]$, where $\tau = |\beta_0 - \beta_{\text{last}}|$.

Advantage. Improves recovery of optimal smooth weights relative to this JS construction.

Limitation. Better recovery of optimal smooth weights; still limited near jumps.

weno5-js

Algorithm. Use fifth-order characteristic FV face candidates with Jiang-Shu weights.

Advantage. Fifth-order candidate blending when smooth stencil data are available.

Limitation. Wider smooth reconstruction; critical-point and boundary effects remain.

weno5-z

Algorithm. Use fifth-order candidates and the Z smoothness correction.

Advantage. Uses a global smoothness difference to reduce unnecessary weight distortion.

Limitation. Adds a global smoothness correction but still requires a wider stencil and admissibility control; improved weights do not guarantee better discontinuous profiles.

17B / WORKED WEIGHTS

Why a troubled stencil loses influence

This synthetic arithmetic example illustrates JS weighting. It is not a fitted Euler stencil or an accuracy experiment. Let candidate face values be $q=(1,3,10)$, optimal weights $d=(0.1,0.6,0.3)$, indicators $\beta=(0.01,0.02,1)$ and $\varepsilon=10^{-6}$.

Key equation

$$\alpha_k = d_k / (\beta_k + \varepsilon)^2, \quad \omega_k = \alpha_k / \sum_j \alpha_j$$

Candidate	q	β	d	α	ω
0	1	0.01	0.1	999.80003	0.3999280
1	3	0.02	0.6	1499.85001	0.5999520
2	10	1	0.3	0.30000	0.0001200

Linear combination: $\sum d q = 4.9000$.

Nonlinear combination: $\sum \omega q = 2.200984$. The large indicator on candidate 2 makes its weight very small.

Smooth-region limit

When all candidate smoothness indicators are approximately equal, the normalized JS weights approach the optimal linear weights d . When a candidate crosses a jump, the indicator should suppress it. Near critical points, the route back to optimal weights can be imperfect; Z weighting addresses smooth-weight recovery differently.

Teaching idea: explain reconstruction, smoothness indicators and normalization as separate operations, following Shu, NASA/CR-97-206253, Sections 2.1-2.3. The numerical example is newly constructed for this lecture; actual code ε settings remain as documented.

WENO7 and targeted TENO

weno7-js

Algorithm. Use seventh-order characteristic FV face candidates and Jiang-Shu weights.

Advantage. Raises formal face accuracy on a sufficiently smooth wider stencil.

Limitation. Higher formal face order and wider stencil; no WENO7-Z option is implemented.

teno3

Algorithm. Normalize inverse-smoothness indicators, discard stencils below a cutoff, then renormalize the optimal linear weights over retained stencils.

Advantage. Kept stencils recover renormalized optimal weights instead of continuously altered weights.

Limitation. Binary selection depends on the detector and threshold. This educational variant has no blanket accuracy or positivity guarantee.

teno5

Algorithm. Use a fifth-order TENO smoothness detector based on τ_5 , a binary stencil cutoff, and retained optimal weights.

Advantage. Targeted detection rejects troubled fifth-order candidate stencils.

Limitation. Binary stencil rejection depends on cutoff and smoothness data; positivity control and adequate boundary treatment are still required.

teno7

Algorithm. Apply inverse-smoothness binary cutoff to seventh-order characteristic FV candidates.

Advantage. Tests targeted selection on a wider, higher-order face stencil.

Limitation. Wider stencils and binary cutoff decisions add sensitivity and work. This educational variant is not TENO-A/LAD.

TENO cutoff= 10^{-5} ; exponent=6. TENO3/7 are educational inverse-smoothness cutoff variants. No TENO-A/LAD or WENO7-Z option is implemented.

THINC and boundary variation diminishing

muscl-thinc-bvd

Algorithm. Construct MC-linear and bounded hyperbolic-tangent interface candidates; compare face-jump variation with neighbor candidates; choose the smaller variation component by component. Use interface steepness **1.6** and positivity controls.

Advantage. A bounded nonpolynomial candidate represents part of an interface inside a cell.

Limitation. Single-stage componentwise selection can make imperfect interface choices and needs positivity controls. It is not the full published multistage BVD family.

- Build MC-linear and bounded tanh (THINC) face pairs.
- Apply the implemented monotone-component conditions.
- Compare boundary-jump variation with neighbor candidates.
- Choose the lower variation componentwise, then control face admissibility.

Decision exercise

Linear candidate boundary jumps 0.08 and 0.05 sum to 0.13. THINC jumps 0.03 and 0.04 sum to 0.07. This local comparison selects THINC. The actual code considers neighbor candidates; the arithmetic illustrates the principle.

Roles remain separate

BVD changes face states; the selected Riemann flux still computes transport. This is a single-stage educational selection, not a complete published multistage P4-THINC-BVD implementation.

20 / DG

Evolve a polynomial in each element

Key equation
$$U_h(\xi,t) = \sum_{k=0 \dots p} U_k(t) P_k(\xi), \quad \xi \text{ in } [-1,1]$$

Multiply by basis functions and integrate by parts. Volume quadrature handles physical flux; common numerical fluxes couple element boundaries. Legendre coefficients make modal degrees of freedom explicit.

Key equation
$$\int_K (\partial U_h / \partial t) v \, dx = \int_K F(U_h) v' \, dx - [\text{Fhat } v]_{\partial K}$$

Degree	Stored content	Smooth order / limitation
0	Cell mean	First order; matches first-order FV for matched steps and flux.
1	Mean + linear mode	Second order; limiting reduces order near shocks.
2	Mean + linear + quadratic modes	Third order; extra work and smaller stable time steps.

- All 15 pointwise fluxes available; JST excluded.
- At least $\max(3,p+2)$ Gauss points; CFL restriction includes $1/(2p+1)$.
- Characteristic TVB controls slopes; altered slopes zero higher modes. Recorded TVB parameter=0.
- Sampled positivity scaling reduces higher modes toward the conserved mean.

Consistency check
Degree-0 DG agrees with first-order FV to roundoff for all shared fluxes. This checks their matched limit; it does not establish higher-degree shock accuracy.

Teaching connection: set test function $v=1$, so $v'=0$. The volume term vanishes and the mean evolves from boundary fluxes, recovering FV. This route follows Persson’s Math 228B DG lectures; our code uses a modal Legendre representation.

21 / SAFEGUARDS

Positivity and conservation are different

- Positive density is insufficient: internal energy after kinetic subtraction must give positive pressure.
- Scale toward an admissible mean instead of independently clipping conserved components.
- Reducing DG higher modes preserves the mean.
- An invalid stage mean requires rejecting/reducing the step; reconstruction scaling cannot fix it.

Key equation

$$U_{\text{limited}}(\mathbf{x}) = U + \theta[U_{\text{candidate}}(\mathbf{x}) - U], \quad 0 \leq \theta \leq 1$$

Sampled controls

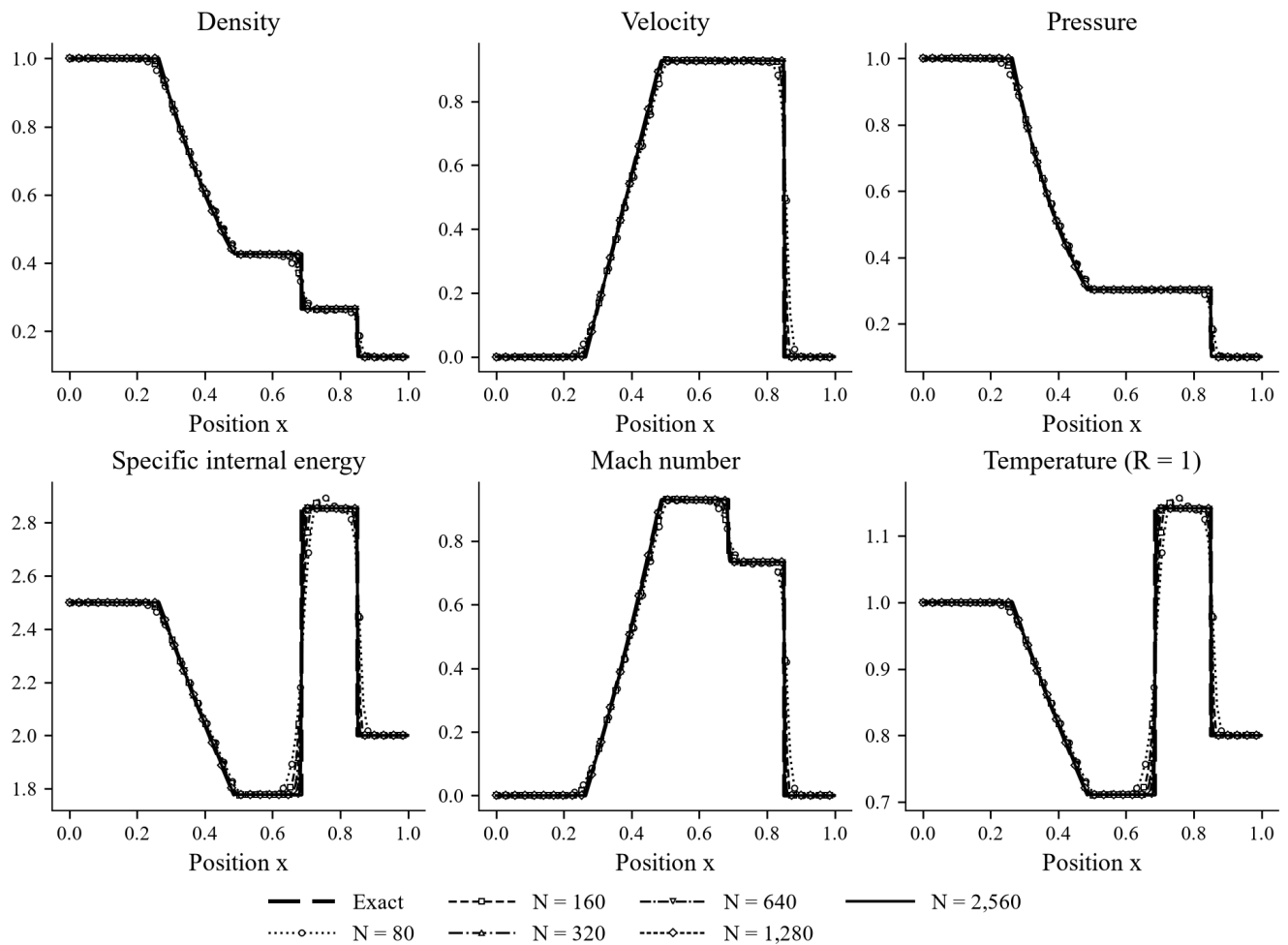
Polynomial/face states are controlled at sample points. Stage means are tested with bounded retry counts. This does not prove admissibility at every point of every polynomial.

Label failures honestly

The original study has 592 runs: 589 completed integrity-passing runs and three earlier-time double-rarefaction failures. Preserve actual stopping times. Do not assign final-time errors to early states.

22 / FLOW PROPERTIES

Read actual computed fields

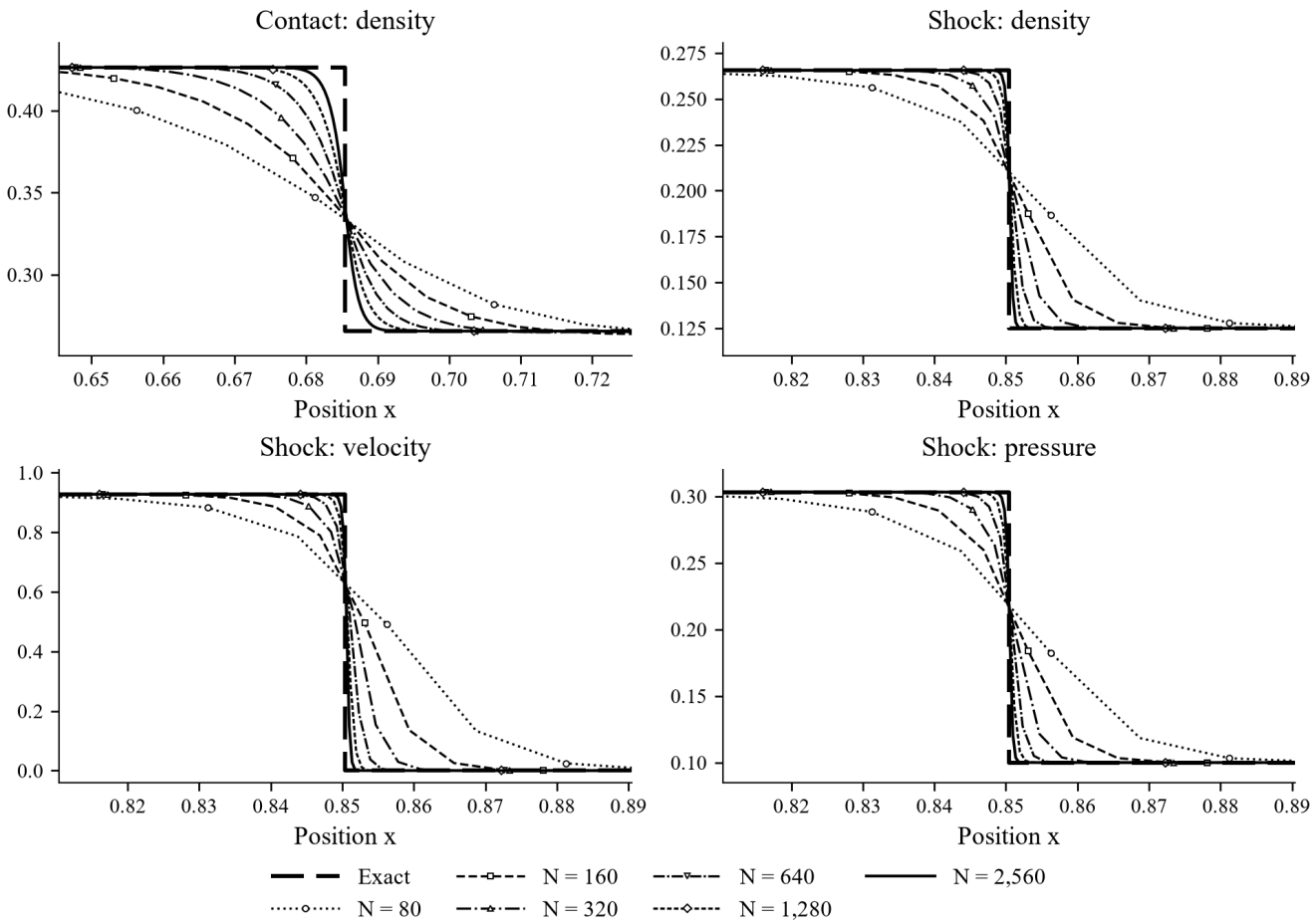


Executed HLLC + CWENO3 at $t=0.2$ on six uniform grids. Curves connect computed cell means; the dashed line is the exact reference.

- Density reveals contact and shock; pressure distinguishes the shock from the contact.
- Derived internal energy, Mach number and temperature come from the same conserved states.
- Keep the physical rarefaction smooth as discontinuous transitions narrow.

With nondimensional $R=1$: $T=p/\rho$; specific internal energy $=p/[(\gamma-1)\rho]$; $Mach=|u|/a$.

Change physical resolution uniformly



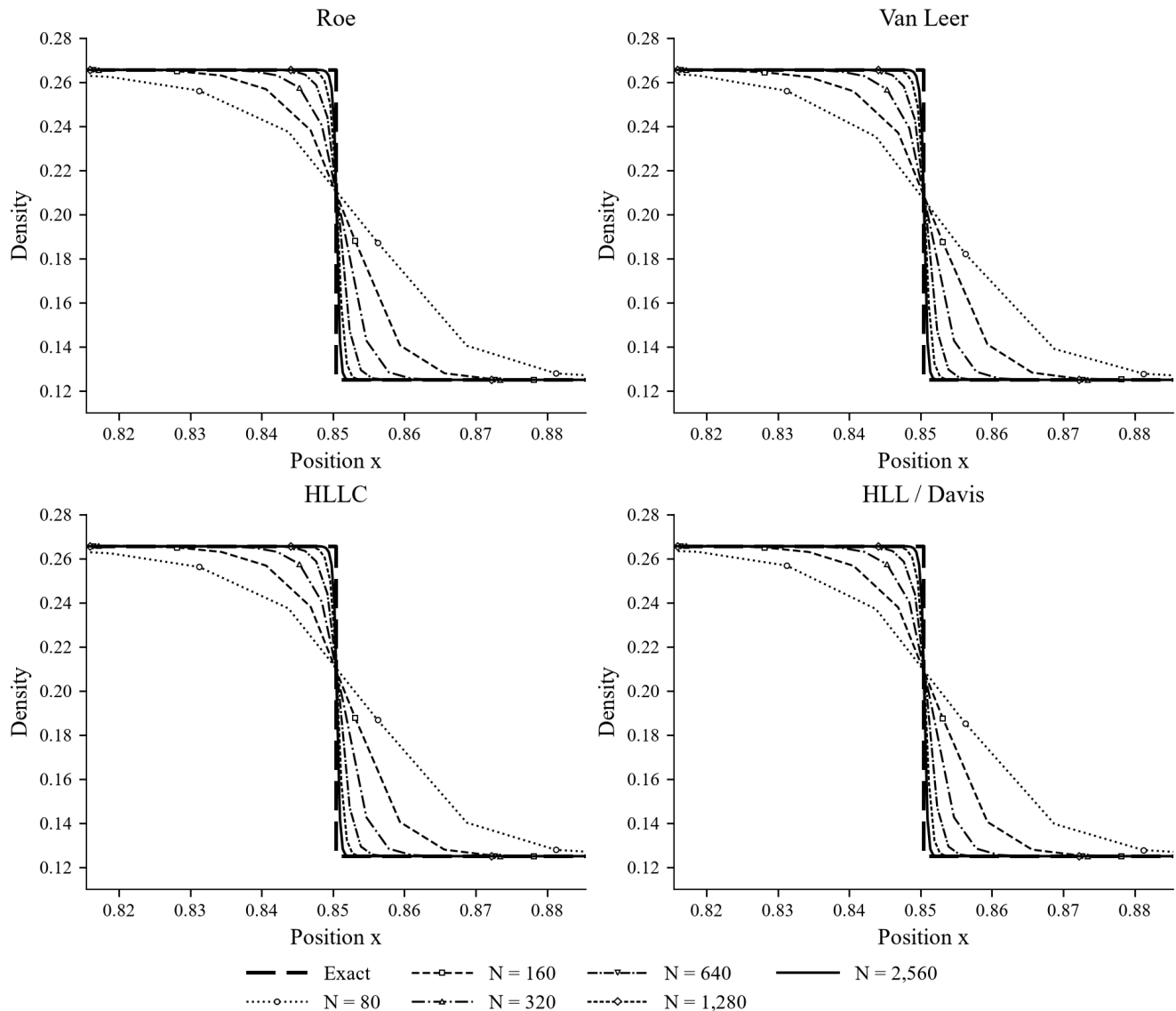
HLLC + CWENO3. N=80,160,320,640,1280,2560; CFL=0.25, t=0.2 fixed. No adaptation or cosmetic sharpening.

N	Density L1	Shock width	Contact width
80	0.00881958	0.0355518	0.0600959
320	0.00239207	0.00906783	0.0225290
1280	0.000694095	0.00225816	0.00807732
2560	0.000416369	0.00112936	0.00481002

HLLC density L1 falls 95.28%. Shock width shrinks about 31.5-fold; contact width 12.5-fold. The finest shock spans 2.89 cells and contact 12.31 cells. Physical narrowing need not mean fewer transition cells.

24 / ALL METHODS / 1 OF 4

Compare waves, not only one error



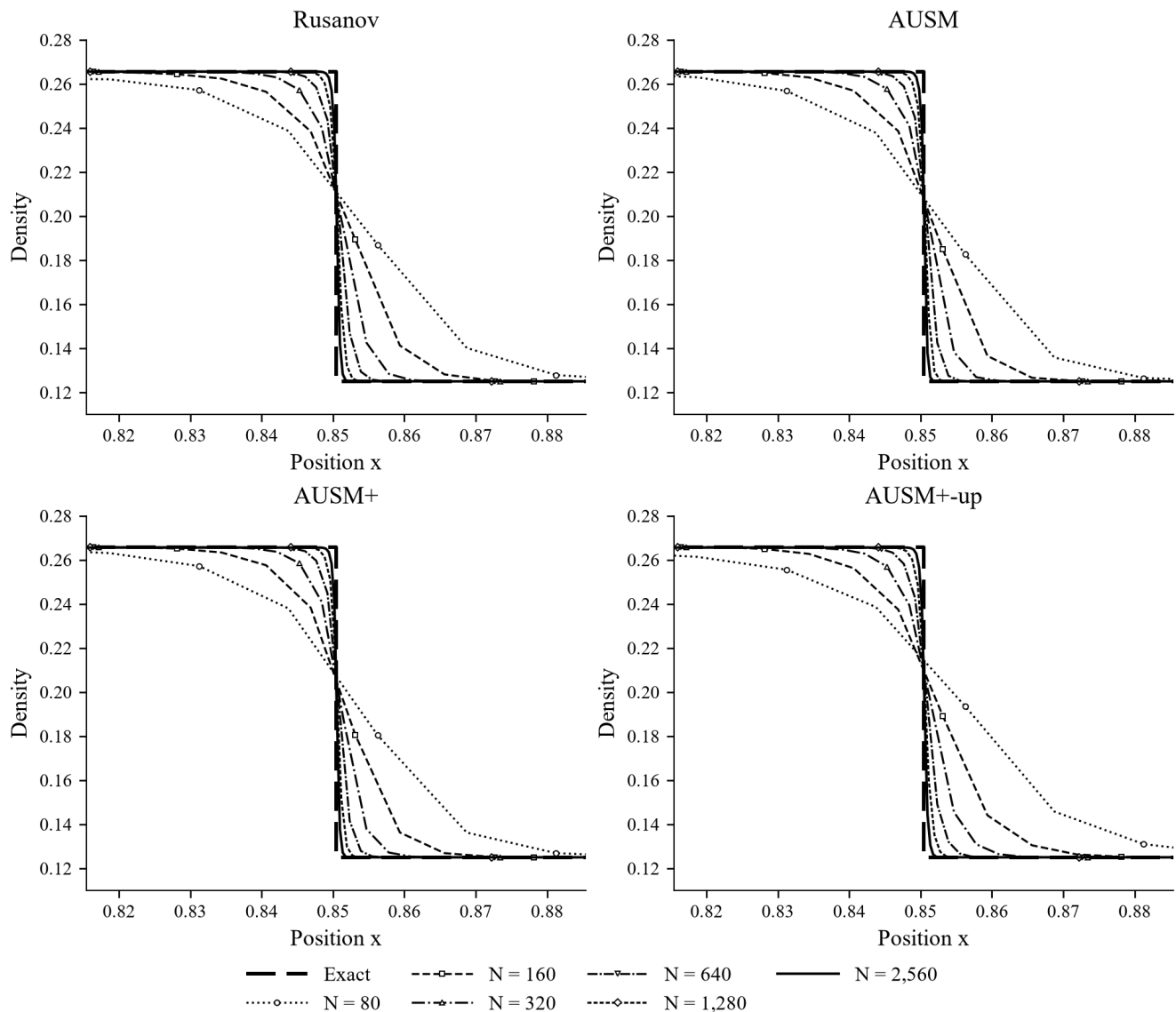
Shock-density detail, group 1. Six uniform grids in each panel; the exact Sod reference is a long-dashed black line.

A controlled flux comparison

All 15 pointwise fluxes use CWENO3; JST uses its own stencil. These four comparison pages keep axis limits, final time and grids fixed. Symbols and line patterns distinguish grids without color.

24 / ALL METHODS / 2 OF 4

Shock resolution: method group 2



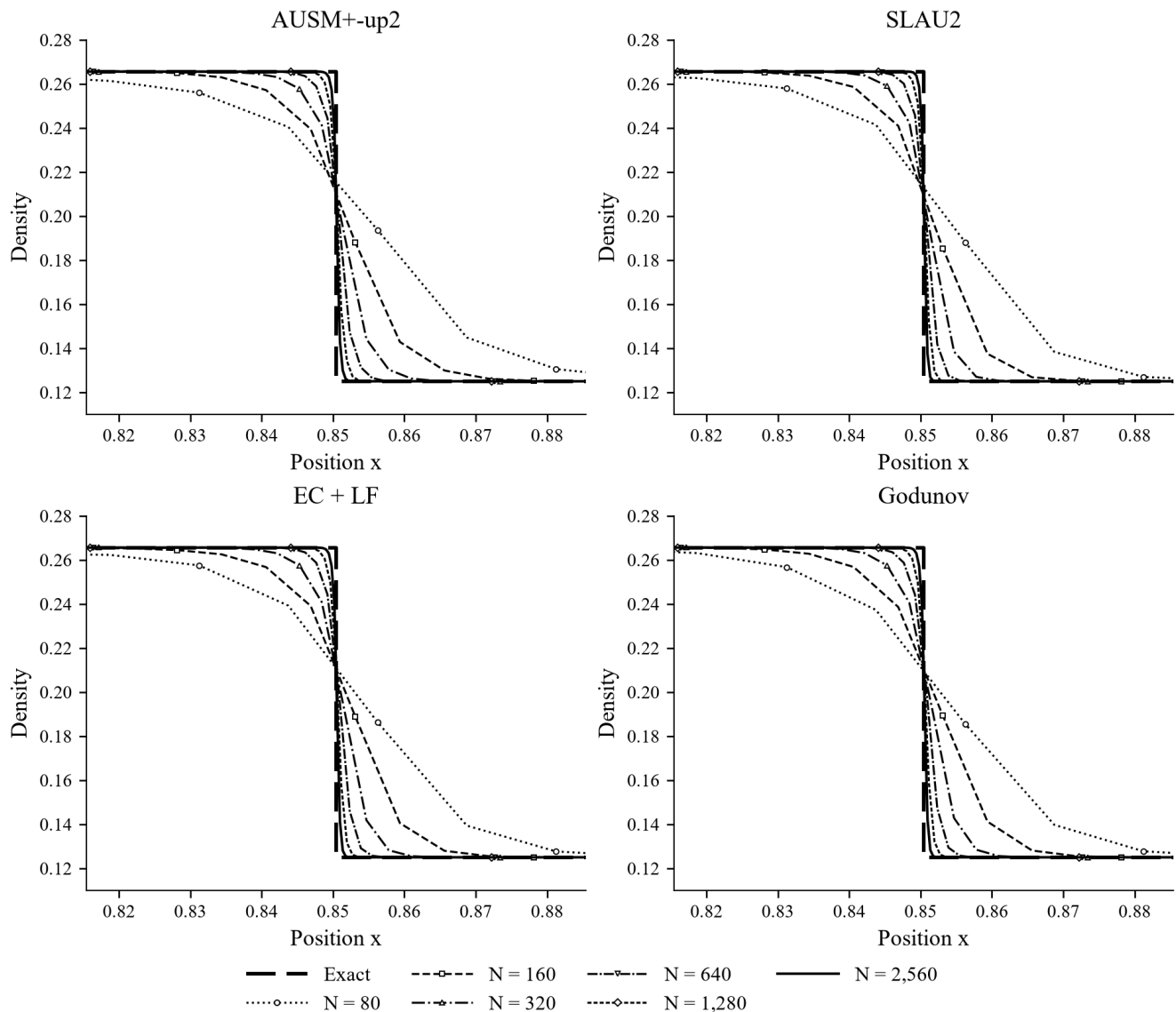
Shock-density detail, group 2. Recorded cell means are connected directly; no numerical smoothing is applied.

Reading prompt

Compare transition position, width and ringing as the uniform grid is refined. A visually narrower transition is useful only when strength, position and exact-reference errors remain consistent.

24 / ALL METHODS / 3 OF 4

Shock resolution: method group 3



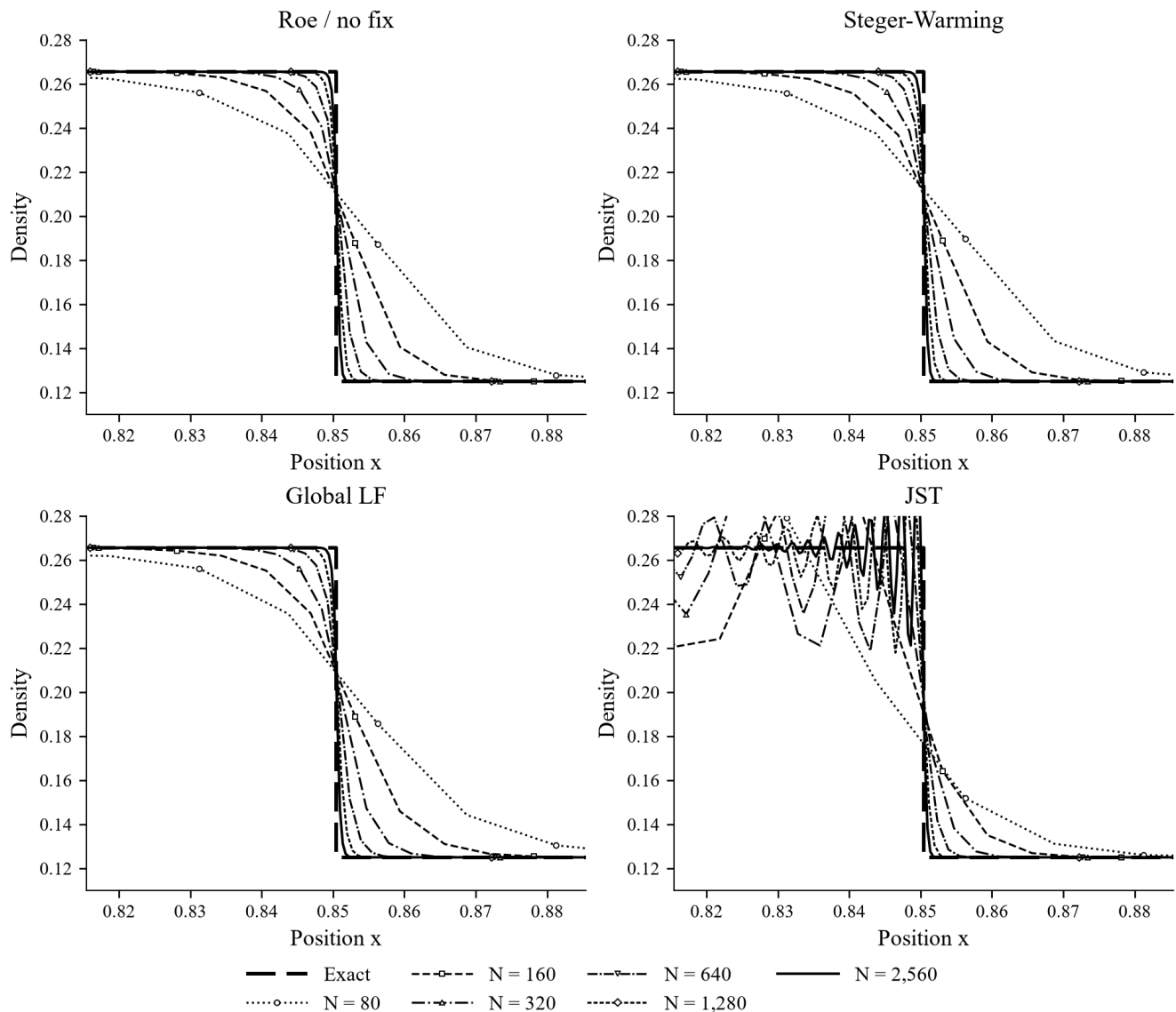
Shock-density detail, group 3. Recorded cell means are connected directly; no numerical smoothing is applied.

Reading prompt

Compare transition position, width and ringing as the uniform grid is refined. A visually narrower transition is useful only when strength, position and exact-reference errors remain consistent.

24 / ALL METHODS / 4 OF 4

Shock resolution: method group 4

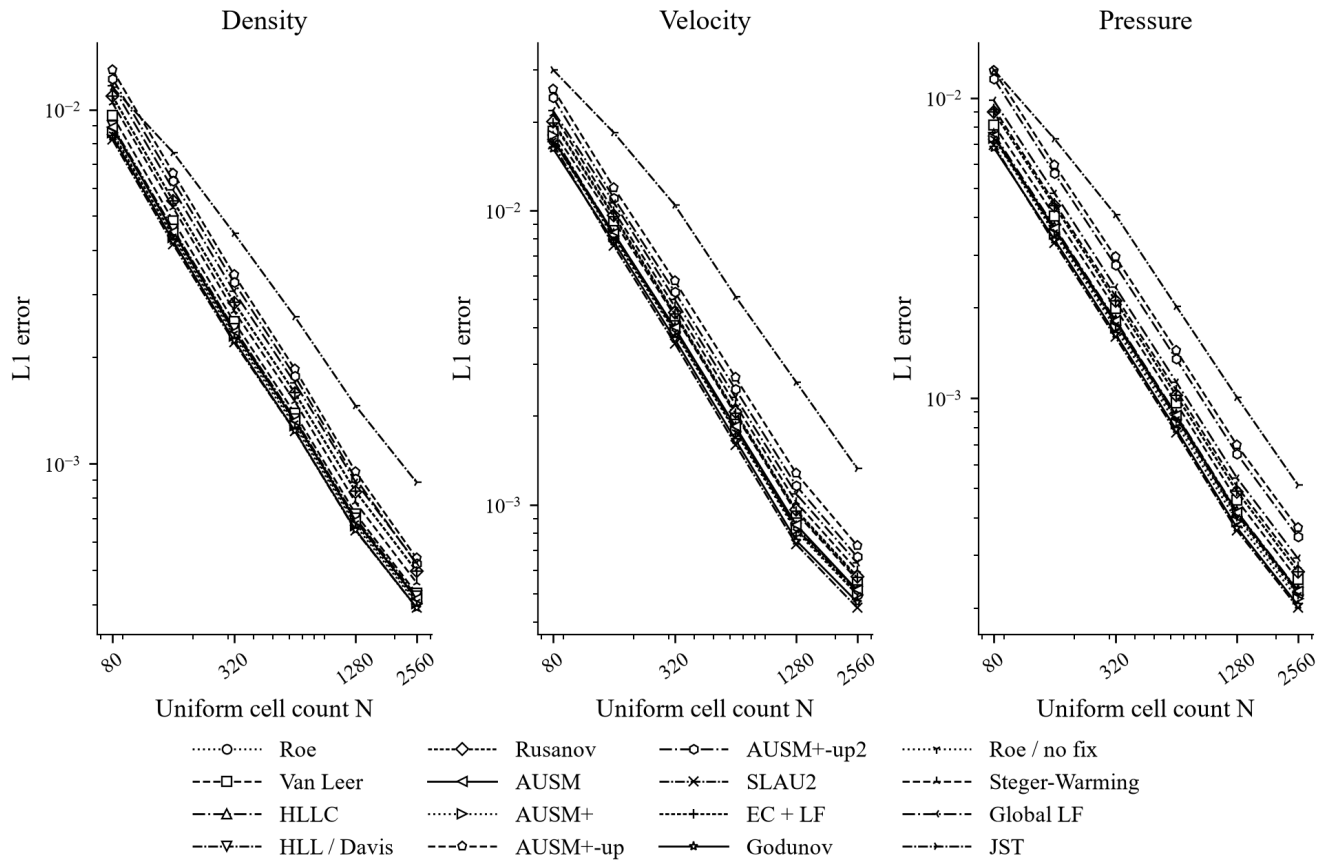


Shock-density detail, group 4. Recorded cell means are connected directly; no numerical smoothing is applied.

An instructive exception

The current JST stencil shows oscillations at the jump. A narrow 10-90% width can miss this weakness. Compare width, overshoot, field errors and admissibility together; preserve unfavorable results too.

Formal order and measured error differ



Actual density, velocity and pressure L1 errors; each algorithm is held fixed as Δx decreases.

Key equation

$$L1(q) = \sum_i |q_{i,num} - q_{i,ref}| \Delta x$$

$$\text{Observed order } r = \log(e_N / e_{2N}) / \log(2)$$

- Integrate exact conserved states into cell means, then convert to primitive variables for the reference.
- Discontinuities lower global observed order even for high-order smooth reconstruction.
- Separate mesh-error convergence from transient RHS/time histories; this is not a steady residual solve.

Actual coverage

96 comparisons = 16 methods $\times$ 6 grids, all integrity-passing. The original full FV matrix = 15 fluxes $\times$ 20 reconstructions = 300 runs on 80 cells. Six-grid refinement does not repeat all 300 combinations or DG.

26 / LABORATORY

Choose methods through controlled tests

Question	Hold fixed	Change / inspect
Flux effect on contact	CWENO3, grid, time, CFL	HLL/HLLC/LF; density width and errors.
Reconstruction effect	HLLC, grid, time, CFL	First/MUSCL/WENO; errors and overshoot.
Steepening benefit	HLLC, stepper, grid	MC/BVD; width plus errors and failures.
Higher DG degree	Flux and final time	Degree, limiting, time step and error.
Enough resolution?	Algorithm and parameters	Uniform N; physical widths and errors.

Student deliverable

- Six-property plot, exact reference and method/grid legend.
- Shock/contact zoom and L1/transition-width table.
- Uniform-grid errors and observed orders with clearly stated norms.
- Admissibility, conservation and failure table; explain one favorable and one unfavorable result.

Teach a simple baseline first

First-order FV + Rusanov is short enough to debug. HLLC + CWENO3 is a useful controlled comparison here. Add one change at a time; no method is declared universally best.

27 / QUESTIONS

Exercises before reading the answers

- 1. Sum mass updates over three cells. Which flux terms remain?
- 2. $CFL=0.25$, $\Delta x=1/160$, $\max(|u|+a)=2$: find Δt . What changes at $N=320$?
- 3. With $a=0.2$, $b=0.1$, find MC, minmod, Van Leer, Superbee and Van Albada slopes. Find faces for $mean=1.2$.
- 4. Why can HLL diffuse a contact more than HLLC with identical reconstruction?
- 5. Why does an exact Godunov face flux not yield an exact full shock-tube computation?
- 6. Density L1 is 0.00881958 at $N=80$ and 0.00453499 at $N=160$. Estimate observed order.
- 7. Can an oscillatory shock have a narrower 10-90% width and still be worse? Name two additional checks.
- 8. Design a fair WENO-JS versus WENO-Z experiment.
- 9. Why do seventh-order faces not establish seventh-order cone accuracy with a CWENO5 source polynomial?
- 10. Why does degree-0 DG agreement with FV not validate degree-2 shock accuracy?

28 / ANSWERS

Answers and discussion prompts

- 1. Shared interior terms cancel; only boundary fluxes remain. SSPRK conservation uses stage-weighted boundary contributions.
- 2. $\Delta t = 0.25 / (160 \times 2) = 0.00078125$. At $N=320$: 0.000390625.
- 3. $MC=0.15$; $\text{minmod}=0.10$; $\text{Van Leer} \approx 0.13333$; $\text{Superbee}=0.20$; $\text{Van Albada} \approx 0.12$. Face values $= 1.2 \pm s/2$.
- 4. HLL retains outer bounds only. HLLC adds the contact speed and star states.
- 5. Reconstruction, finite cells, time integration, limiting and boundaries still introduce error. Exact refers only to the local Riemann problem.
- 6. $\log(0.00881958/0.00453499)/\log(2) \approx 0.96$. This is not CWENO3 smooth formal order.
- 7. Yes. Check overshoot/undershoot and exact-reference errors, as well as positivity/conservation.
- 8. Hold flux, grid, CFL, time, boundaries and characteristic treatment fixed; change only weights. Then repeat matched grids.
- 9. Source integration participates in the whole spatial operator. Higher face order cannot override a lower-order source component.
- 10. Degree 0 checks the mean update. Degrees 1/2 additionally require volume quadrature, modal evolution, limiting and appropriate time steps.

29 / SOURCES

Connect lessons to runnable code

Open the course code and teaching files

Topic	Shared conical/ module
FV / DG / cases	shock_suite.py and shared DG core
Roe, HLL/C, LF, Van Leer	flux.py
Godunov, uncorrected Roe, SW, JST	additional_fluxes.py
AUSM / SLAU2	ausm.py
Entropy core + LF	entropy_flux.py
CWENO / MUSCL / characteristic weights	reconstruction.py
ENO / WENO / TENO / BVD	advanced_reconstruction.py
Recorded grids and diagnostics	results-uniform/catalog.json; shock-tube/UNIFORM_GRID_RESULTS.md

Primary reading and code comparisons

NASA shocktube: AUSM/AUSM+ examples

Sun, Inaba and Xiao: Boundary Variation Diminishing (2016)

Ihme Group Quail: DG teaching and prototyping

Riemann-Solvers: exact and approximate examples

Descriptions and numerical values follow this course implementation and measured records. External codes/papers are reading references; their full algorithms and guarantees are not automatically implemented here.

Report with every numerical claim

Equations, flux, reconstruction, grid, boundaries, stepper, CFL, parameters, actual stopping time, reference definition and integrity checks. Preserve attribution and distinguish formal smooth order from measured error.

30 / OPEN LECTURE READING MAP

What inspired the teaching structure?

LeVeque / University of Washington

Finite Volume Methods for Hyperbolic Problems: 2023 supplementary lectures and PDF slides

Teaching route: conservation and waves before high-resolution comparisons; pair explanations with runnable experiments.

Ketcheson, LeVeque and del Razo

Riemann Problems and Jupyter Solutions: Euler approximate solvers

Teaching route: local wave fans first, then approximate solvers and their failure mechanisms. The schematic here is newly drawn.

Shu / Brown; NASA ICASE report (1997)

ENO and WENO lecture notes, Sections 2.1-2.3

Teaching route: cell-average reconstruction, candidate stencils and nonlinear weights. The worked weight arithmetic here is new.

Persson / UC Berkeley Math 228B

Discontinuous Galerkin Methods for Conservation Laws

Teaching route: derive FV from piecewise constants, then extend to element polynomials and common boundary fluxes.

Access checked 08 October 2026. These notes use original wording, exercises, diagrams and the course's own measured figures. External lecture PDFs are linked, not reproduced. Their finite-difference, adaptive or viscous extensions are not added to this benchmark.

Michigan context: Fidkowski's official teaching page lists CFD I/II, but that page does not supply a downloadable 2025/2026 lecture set. No such set is claimed as the source of these notes. Official teaching page.